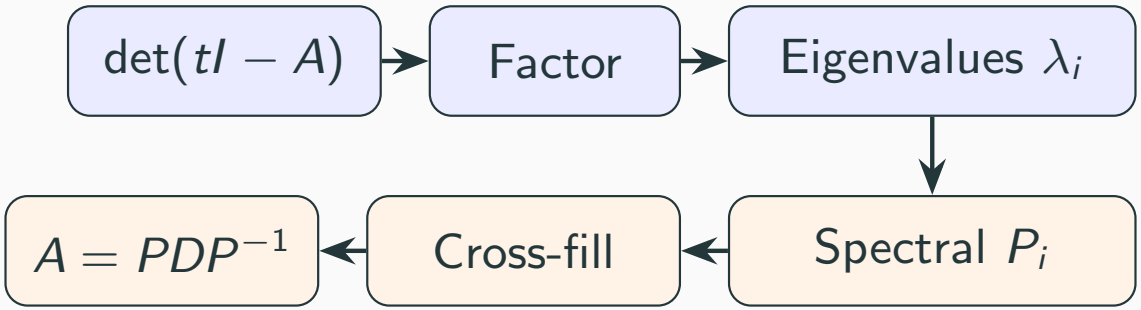
Lecture 17: Complex Numbers

MATH203 Applied Linear Algebra

Our Machinery Breaks Over $\mathbb{R}$

Recall: The Pipeline from L14–L16

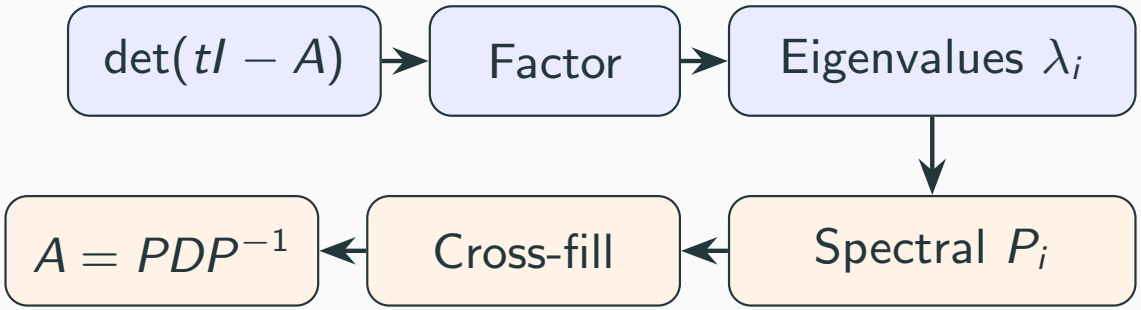
In Lectures 14–16, we built a powerful pipeline:



This works when the minimal polynomial factors into **distinct linear factors**.

Recall: The Pipeline from L14–L16

In Lectures 14–16, we built a powerful pipeline:



This works when the minimal polynomial factors into **distinct linear factors**.

But there are **two restrictions** in this assumption:

(a) The polynomial must factor into **linear** factors.

(b) Those factors must be **distinct** (no repeated roots in the minimal polynomial).

Today and Beyond

Restriction	Problem	Solution
(a) Cannot factor into linear factors over $\mathbb{R}$	Some polynomials are irreducible over $\mathbb{R}$	**Today:** Complex numbers $\mathbb{C}$
(b) Repeated factors in the minimal polynomial	Lagrange interpolation fails: observation points collapse, losing information	Self-study: Jordan Canonical Form

Today and Beyond

Restriction	Problem	Solution
(a) Cannot factor into linear factors over $\mathbb{R}$	Some polynomials are irreducible over $\mathbb{R}$	**Today:** Complex numbers $\mathbb{C}$
(b) Repeated factors in the minimal polynomial	Lagrange interpolation fails: observation points collapse, losing information	Self-study: Jordan Canonical Form

Today we address **restriction (a)** only.

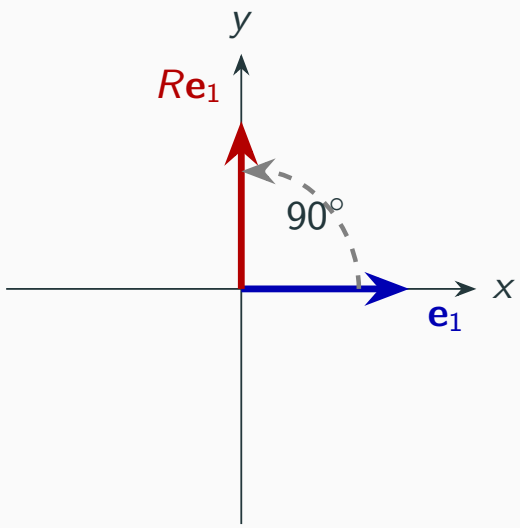
Restriction (b) requires completely different tools (Jordan Canonical Form) — this will be **self-study material**.

Complex numbers do **not** solve restriction (b).

The Pipeline Breaks: A Concrete Example

The **90° rotation matrix**:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

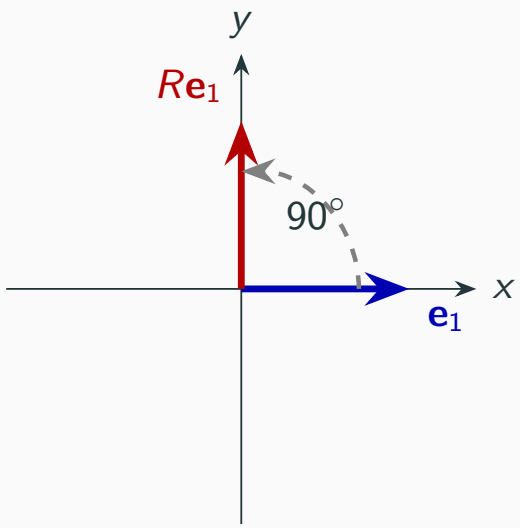


We know **exactly** what R does: it rotates every vector by 90° counterclockwise. Geometrically clear. No mystery.

The Pipeline Breaks: A Concrete Example

The **90° rotation matrix**:

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$



We know **exactly** what R does: it rotates every vector by 90° counterclockwise. Geometrically clear. No mystery.

Compute the characteristic polynomial:

$$\det(tI - R) = \det \begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix} = t^2 + 1$$

The Pipeline Stops

$$\det(tI - R) = t^2 + 1$$

$t^2 + 1$ has **no real roots**.

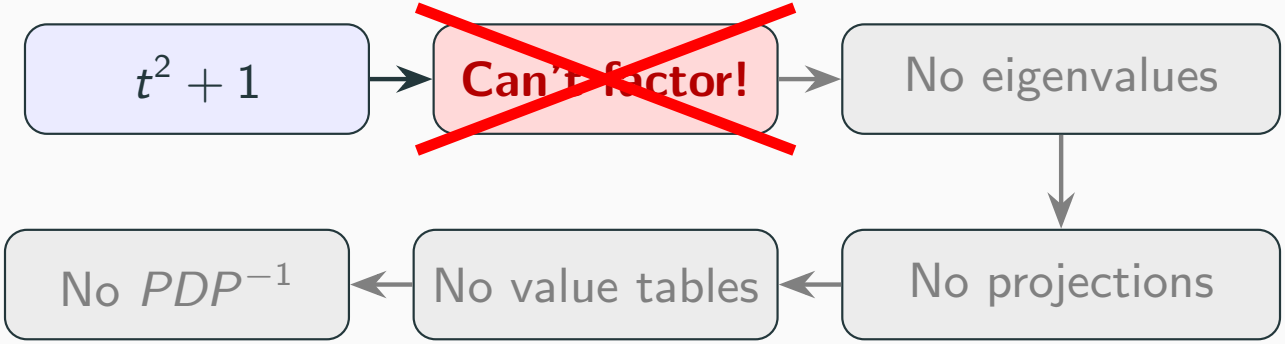
It does **not** factor into linear factors over $\mathbb{R}$.

The Pipeline Stops

$$\det(tI - R) = t^2 + 1$$

$t^2 + 1$ has **no real roots**.

It does **not** factor into linear factors over $\mathbb{R}$.



The matrix R is **perfectly concrete** — it rotates by $90°$.

The problem is not the matrix. The problem is the **number system**.

The Question

Can we extend $\mathbb{R}$ to a **larger number system** where $t^2 + 1 = 0$ **has** solutions?

The Question

Can we extend $\mathbb{R}$ to a **larger number system** where $t^2 + 1 = 0$ **has** solutions?

And not just for this one polynomial — does **every** polynomial factor into linear factors there?

The Question

Can we extend $\mathbb{R}$ to a **larger number system** where $t^2 + 1 = 0$ **has** solutions?

And not just for this one polynomial — does **every** polynomial factor into linear factors there?

Spoiler: Fundamental Theorem of Algebra

Over the **complex numbers** $\mathbb{C}$, every degree-n polynomial has exactly n roots (counted with multiplicity).

$$p(t) = (t - z_1)(t - z_2) \cdots (t - z_n), \qquad z_i \in \mathbb{C}$$

Every polynomial factors into linear factors. Restriction (a) is completely removed.

The Cubic Formula: Complex Numbers Are Unavoidable

The Quadratic Formula

For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

When $b^2 - 4ac < 0$: "no real solutions."

The Quadratic Formula

For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

When $b^2 - 4ac < 0$: "no real solutions."

Easy to dismiss $\sqrt{-1}$ — just say "no solution" and move on.

But what about **cubic** equations?

The Cubic Equation

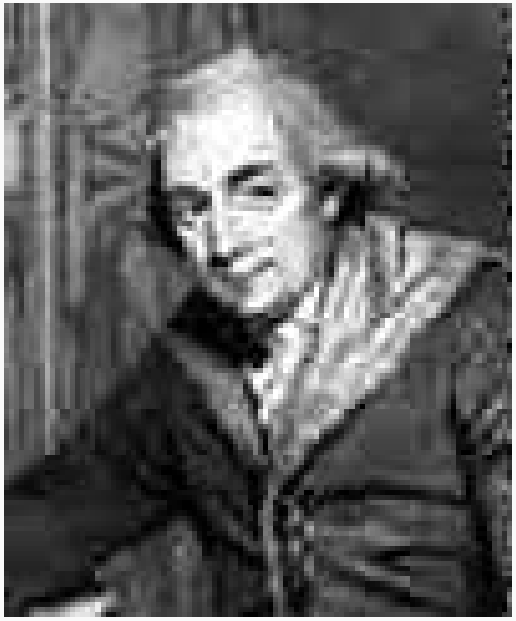
$$ax^3 + bx^2 + cx + d = 0$$

Unsolved for centuries — until **1530**.
Scipione del Ferro found a method, kept it secret until his deathbed. His student Antonio Fior learned it. Eventually Cardano published the formula.

The Cubic Equation

$$ax^3 + bx^2 + cx + d = 0$$

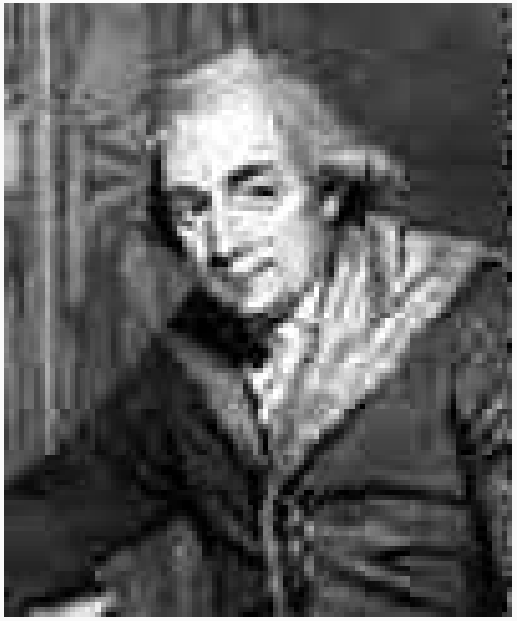
Unsolved for centuries — until **1530**.
Scipione del Ferro found a method, kept it secret until his deathbed. His student Antonio Fior learned it. Eventually Cardano published the formula.
Why doesn't your teacher show you this formula?



The Cubic Equation

$$ax^3 + bx^2 + cx + d = 0$$

Unsolved for centuries — until **1530**.
Scipione del Ferro found a method, kept it secret until his deathbed. His student Antonio Fior learned it. Eventually Cardano published the formula.
Why doesn't your teacher show you this formula?



Because it looks like this:

$$\boxed{x = \sqrt[3]{\left(-\frac{b^3}{27a^2} + \frac{bc}{6a^2} - \frac{d}{2a}\right) + \sqrt{\Delta}} + \sqrt[3]{\left(-\frac{b^3}{27a^2} + \frac{bc}{6a^2} - \frac{d}{2a}\right) - \sqrt{\Delta}} - \frac{b}{3a}}$$

where $\Delta = \left(-\frac{b^3}{27a^2} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3$.

The Paradox

Use the cubic formula on a simple equation:

$$x^3 = x$$

Obviously the solutions are $x = 0,\ 1,\ -1$.

The Paradox

Use the cubic formula on a simple equation:

$$x^3 = x$$

Obviously the solutions are $x = 0,\ 1,\ -1$.

But the cubic formula gives:

$$x = \frac{1}{\sqrt{3}}\left(\sqrt[3]{\sqrt{-1}} + \sqrt[3]{-\sqrt{-1}}\right)$$

The Paradox

Use the cubic formula on a simple equation:

$$x^3 = x$$

Obviously the solutions are $x = 0,\ 1,\ -1$.

But the cubic formula gives:

$$x = \frac{1}{\sqrt{3}}\left(\sqrt[3]{\sqrt{-1}} + \sqrt[3]{-\sqrt{-1}}\right)$$

This looks **insane**. There is a $\sqrt{-1}$ inside a cube root!

And yet... if you compute with $\sqrt{-1}$ formally, you get back $0,\ 1,\ -1$.

Bombelli's Insight (1572)

Rafael Bombelli was the first to take $\sqrt{-1}$ seriously.

His discovery: if you **compute with $\sqrt{-1}$ formally** — treating it as an algebraic symbol with $(\sqrt{-1})^2 = -1$ — the cubic formula **works**.

Bombelli's Insight (1572)

Rafael Bombelli was the first to take $\sqrt{-1}$ seriously.

His discovery: if you **compute with $\sqrt{-1}$ formally** — treating it as an algebraic symbol with $(\sqrt{-1})^2 = -1$ — the cubic formula **works**.

$\sqrt{-1}$ appears in the **middle** of the calculation...

...but **cancels out** at the end, giving real answers.

Bombelli's Insight (1572)

Rafael Bombelli was the first to take $\sqrt{-1}$ seriously.

His discovery: if you **compute with $\sqrt{-1}$ formally** — treating it as an algebraic symbol with $(\sqrt{-1})^2 = -1$ — the cubic formula **works**.

$\sqrt{-1}$ appears in the **middle** of the calculation...

...but **cancels out** at the end, giving real answers.

For centuries, mathematicians had assumed $\sqrt{-1}$ "does not exist."

But this assumption **blocked** them from using a perfectly valid formula.

You cannot assume $\sqrt{-1}$ does not exist. It is meaningful and must be computed.

The Lesson

Complex numbers are not optional

Even when the **problem** involves only real numbers and the **answer** is real, the **solution path** may force you through $\sqrt{-1}$.

This is not a trick or a convenience — it is a **mathematical necessity**.

The Lesson

Complex numbers are not optional

Even when the **problem** involves only real numbers and the **answer** is real, the **solution path** may force you through $\sqrt{-1}$.

This is not a trick or a convenience — it is a **mathematical necessity**.

This pattern appears throughout science:

- **Waves:** oscillations naturally described by $e^{i\theta}$
- **Quantum mechanics:** the Schrödinger equation lives in $\mathbb{C}$
- **Electrical engineering:** impedance uses complex arithmetic

Even when the physical measurement is real, the **mathematics behind it** uses complex numbers.

The Complex Numbers

Definition

We write $i = \sqrt{-1}$, so that $i^2 = -1$.

Definition

A **complex number** is a number of the form

$$z = a + bi, \qquad a, b \in \mathbb{R}$$

where a is the **real part** and b is the **imaginary part**.

The set of all complex numbers is denoted $\mathbb{C}$.

Definition

We write $i = \sqrt{-1}$, so that $i^2 = -1$.

Definition

A **complex number** is a number of the form

$$z = a + bi, \qquad a, b \in \mathbb{R}$$

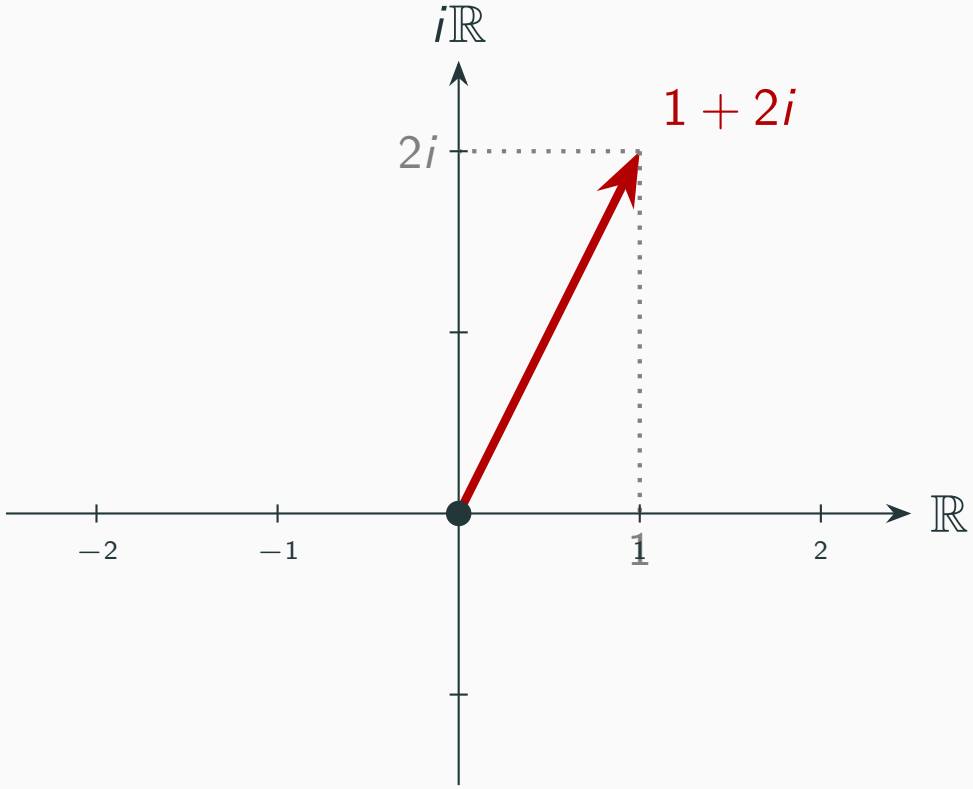
where a is the **real part** and b is the **imaginary part**.

The set of all complex numbers is denoted $\mathbb{C}$.

$$\underbrace{\mathbb{N}}_{\text{natural}} \subset \underbrace{\mathbb{Z}}_{\text{integer}} \subset \underbrace{\mathbb{Q}}_{\text{rational}} \subset \underbrace{\mathbb{R}}_{\text{real}} \subset \underbrace{\mathbb{C}}_{\text{complex}}$$

The Complex Plane

A complex number $a + bi$ corresponds to a point (a, b) in $\mathbb{R}^2$.



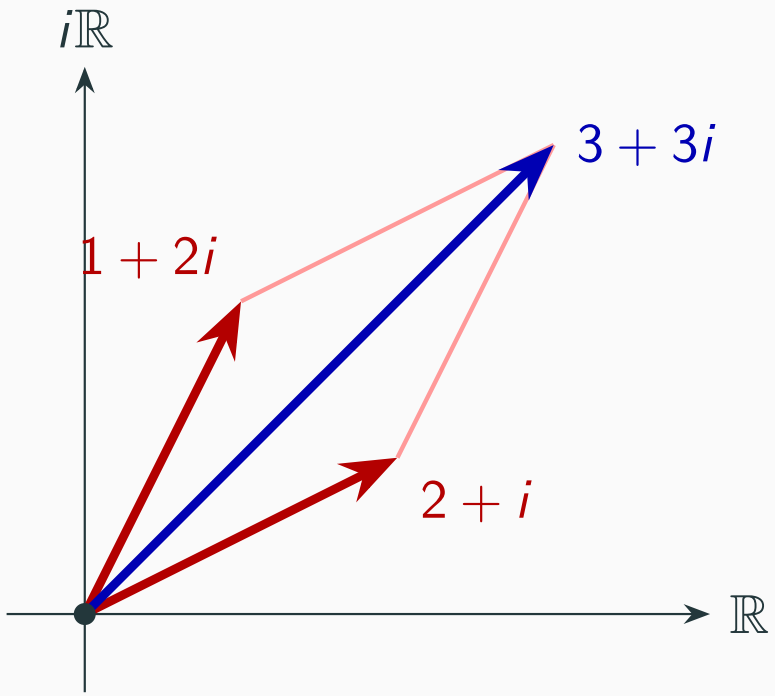
The horizontal axis is the **real axis** ($\mathbb{R}$).

The vertical axis is the **imaginary axis** ($i\mathbb{R}$).

Addition = Parallelogram Law

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

Addition of complex numbers is **the same as vector addition in $\mathbb{R}^2$**.



$(1 + 2i) + (2 + i) = 3 + 3i$

Multiplication: The Surprise

Addition is familiar — same as vectors.

But complex numbers can be **multiplied**, and this is where the magic happens.

Multiplication: The Surprise

Addition is familiar — same as vectors.

But complex numbers can be **multiplied**, and this is where the magic happens.

To multiply, expand as a polynomial in i and use $i^2 = -1$:

$$(3+i)(5+i) = 15 + 3i + 5i + \underbrace{i^2}_{=-1} = 14 + 8i$$

Multiplication: The Surprise

Addition is familiar — same as vectors.

But complex numbers can be **multiplied**, and this is where the magic happens.

To multiply, expand as a polynomial in i and use $i^2 = -1$:

$$(3+i)(5+i) = 15 + 3i + 5i + \underbrace{i^2}_{=-1} = 14 + 8i$$

In general:

$$(a+bi)(c+di) = \underbrace{(ac-bd)}_{\text{real part}} + \underbrace{(ad+bc)}_{\text{imaginary part}} \cdot i$$

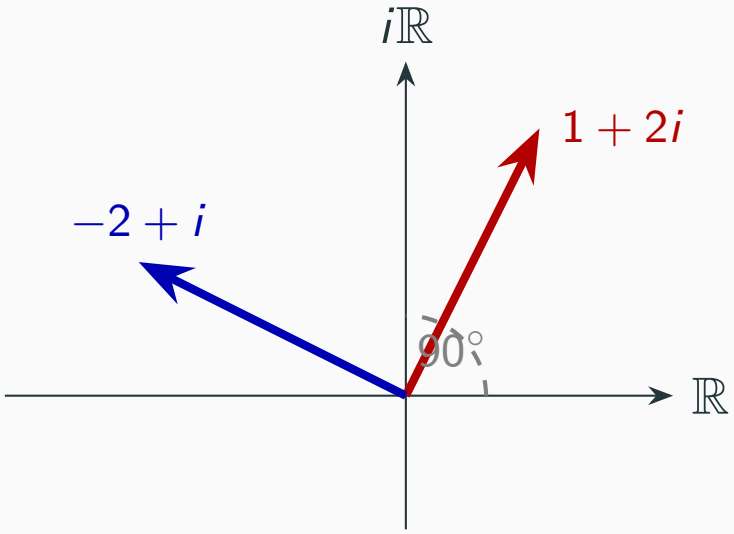
Next section: What does multiplication **look like** geometrically?

Multiplication = Rotation + Scaling

Multiplying by i

Since $i^2 = -1$:

$$i \cdot (a + bi) = -b + ai$$

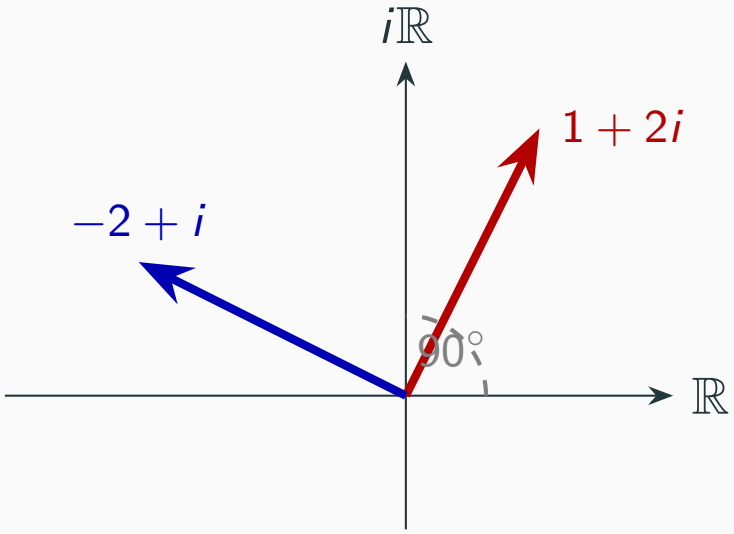


$$i \cdot (1 + 2i) = -2 + i$$

Multiplying by i

Since $i^2 = -1$:

$$i \cdot (a + bi) = -b + ai$$



$i \cdot (1 + 2i) = -2 + i$

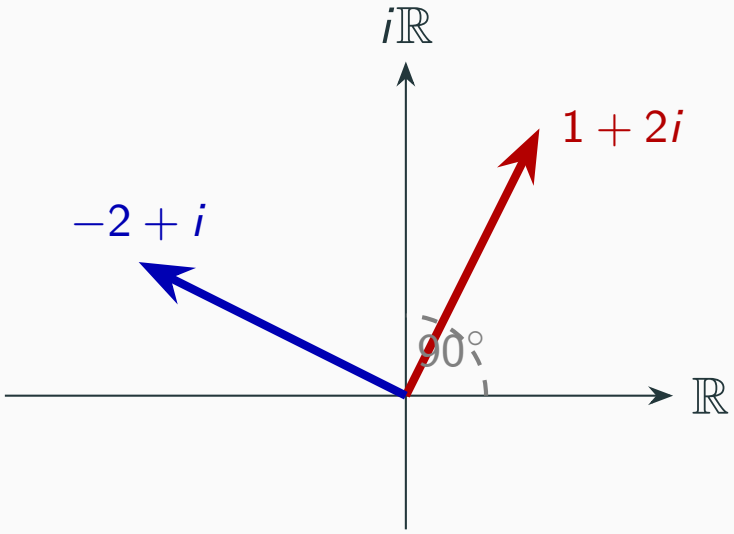
Multiplying by i rotates by $90°$ counterclockwise!

No scaling (the length stays the same).

Multiplying by i

Since $i^2 = -1$:

$$i \cdot (a + bi) = -b + ai$$



$i \cdot (1 + 2i) = -2 + i$

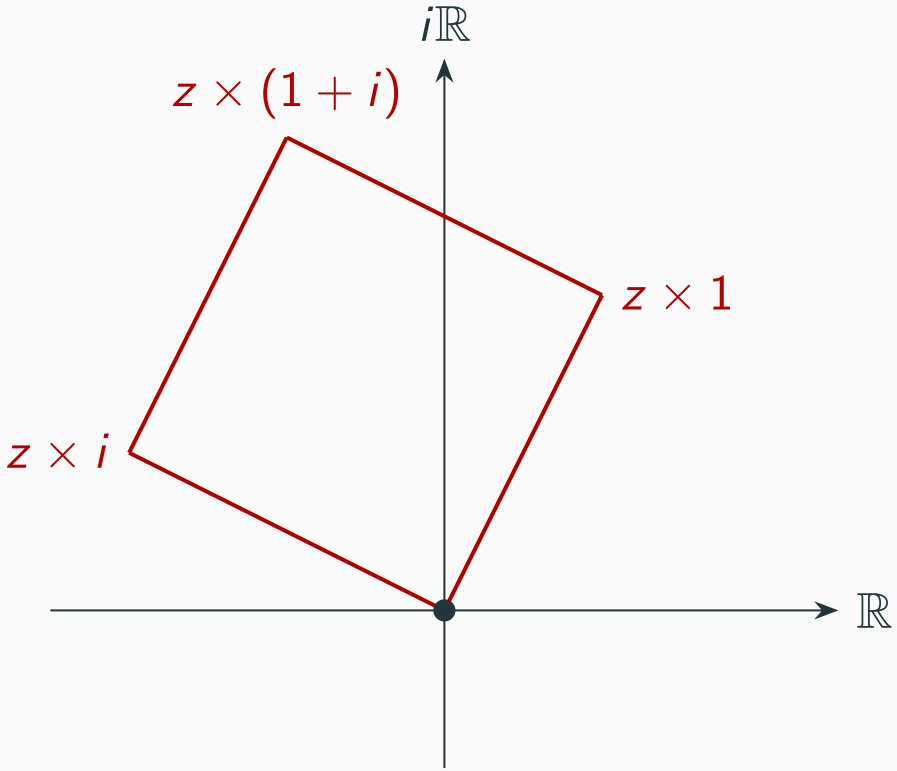
Multiplying by i rotates by $90°$ counterclockwise!

No scaling (the length stays the same).

Connection to §1: The rotation matrix $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ maps $\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} -b \\ a \end{pmatrix}$ — exactly multiplication by i.

The Square Construction

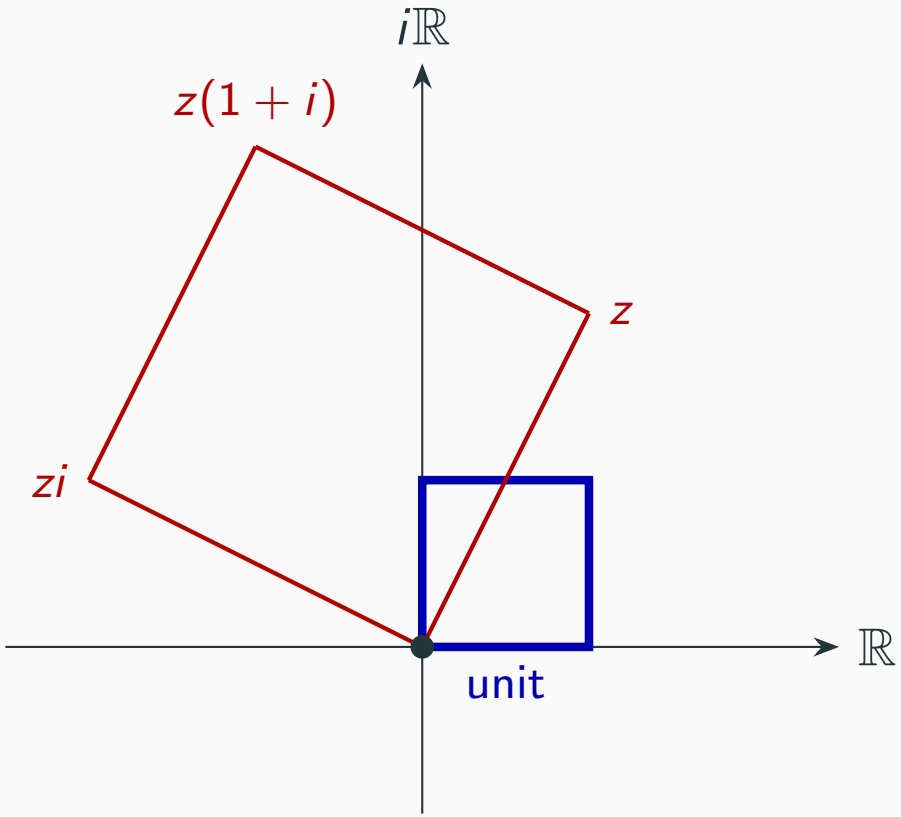
For any complex number z, the three points z, zi, and $z(1+i)$ form a **square** with the origin.



Here $z = 1 + 2i$. Since zi is always z rotated by $90°$, the four points $0, z, zi, z(1+i)$ form a square.

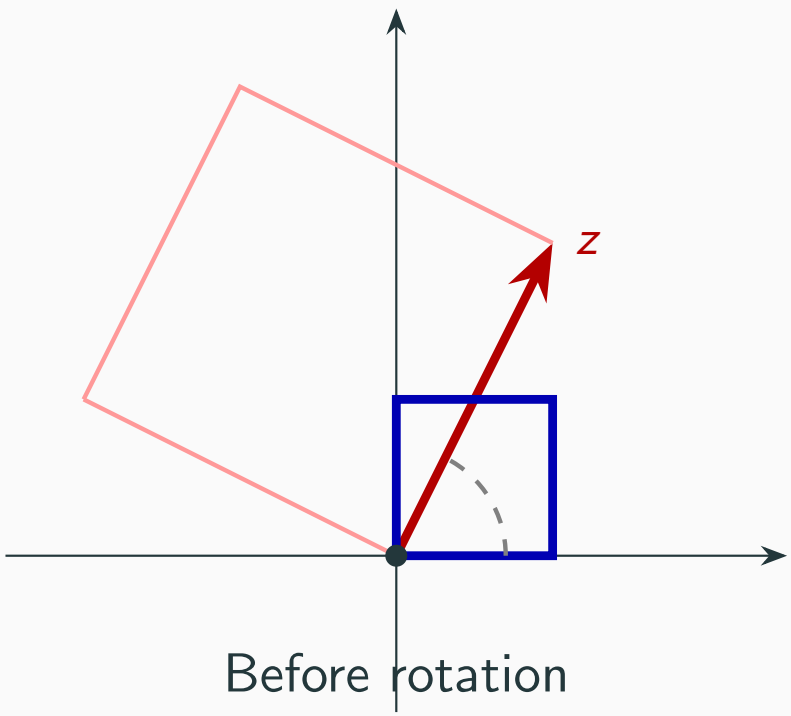
What Does "Multiply by z" Do?

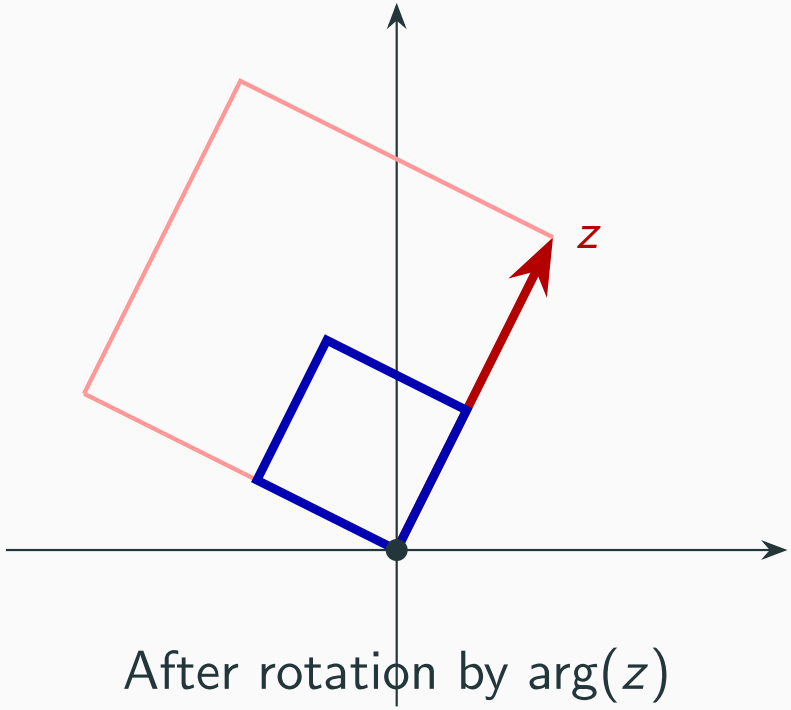
Multiplying by z sends the blue unit square $\{0, 1, i, 1+i\}$ to the red square $\{0, z, zi, z(1+i)\}$:



Here $z = 1 + 2i$. The blue 1×1 square is mapped to the red tilted square.

Step 1: Rotate

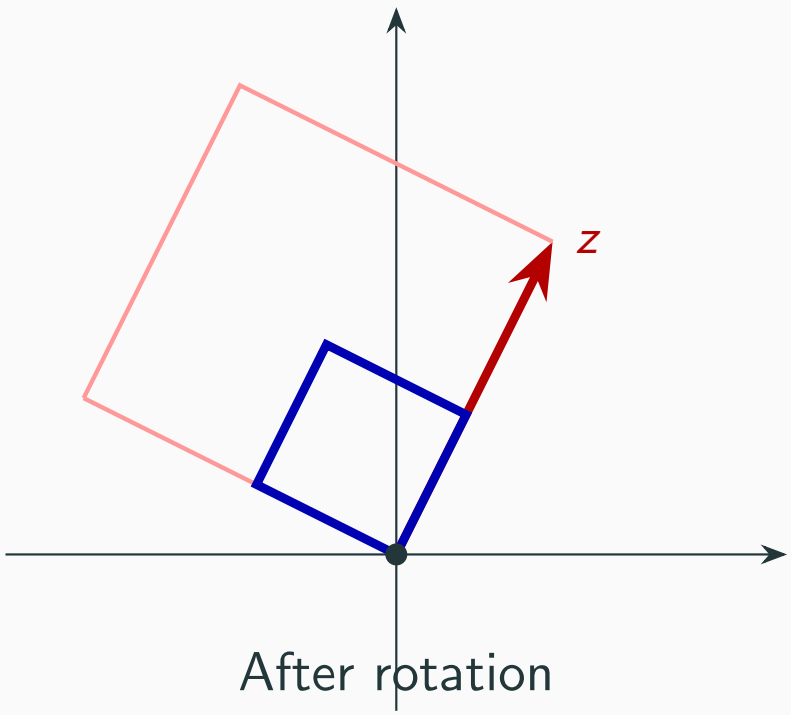


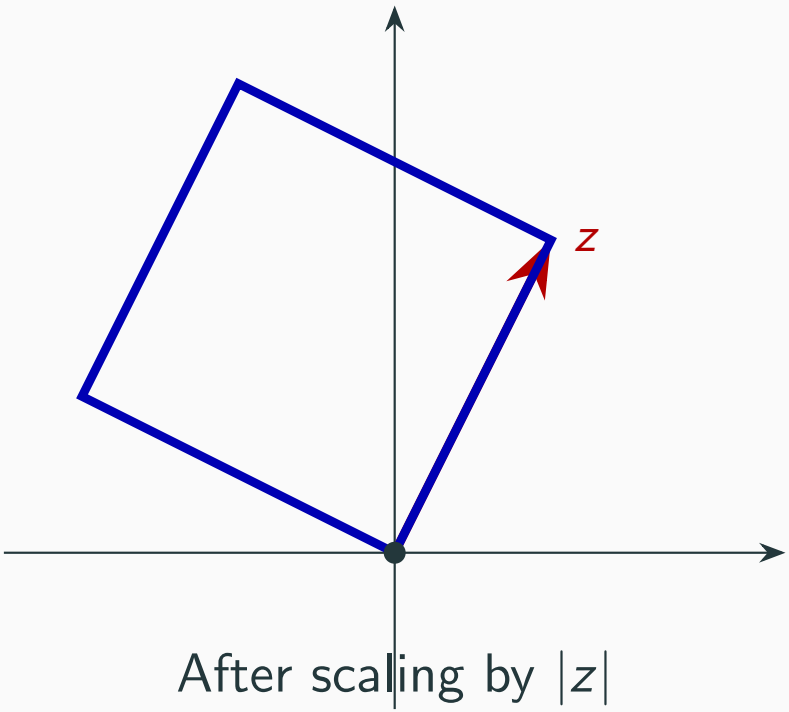


Rotate the blue square by the angle $\arg(z) = \arctan \frac{b}{a}$.

This is the angle between z and the positive real axis.

Step 2: Scale





Scale by $|z| = \sqrt{a^2 + b^2}$, the **absolute value** of z.

For $z = 1 + 2i$: $|z| = \sqrt{1^2 + 2^2} = \sqrt{5}$.

Summary: Multiplication = Rotation + Scaling

Geometric meaning of complex multiplication

Multiplying by $z = a + bi$:

1. **Rotate** by $\arg(z) = \arctan \frac{b}{a}$ (the angle of z)
2. **Scale** by $|z| = \sqrt{a^2 + b^2}$ (the length of z)

Equivalently: the unique transformation that sends $1 \mapsto z$.

Summary: Multiplication = Rotation + Scaling

Geometric meaning of complex multiplication

Multiplying by $z = a + bi$:

1. **Rotate** by $\arg(z) = \arctan \frac{b}{a}$ (the angle of z)
2. **Scale** by $|z| = \sqrt{a^2 + b^2}$ (the length of z)

Equivalently: the unique transformation that sends $1 \mapsto z$.

Special case: Multiplying by i:

- $|i| = 1$ (no scaling)
- $\arg(i) = 90^\circ$ (pure rotation)

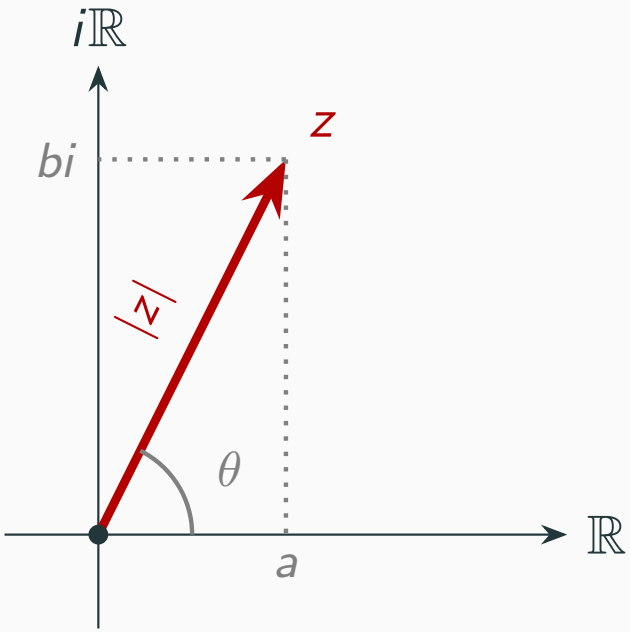
This is why multiplication by i is exactly 90° rotation.

Polar Form, De Moivre & Roots of Unity

Polar Form

From §4, every complex number $z = a + bi$ is determined by:

- its **absolute value** $|z| = \sqrt{a^2 + b^2}$ (distance from origin)
- its **argument** $\arg(z) = \arctan \frac{b}{a}$ (angle from real axis)



Polar form:

$$z = |z|(\cos\theta + i\sin\theta)$$

where $\theta = \arg(z)$.
Real part: $a = |z|\cos\theta$
Imaginary part: $b = |z|\sin\theta$

Multiplication in Polar Form

If $z_1 = |z_1|(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = |z_2|(\cos\theta_2 + i\sin\theta_2)$, then:

$$z_1 z_2 = |z_1||z_2|\big(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)\big)$$

Multiplication in Polar Form

If $z_1 = |z_1|(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = |z_2|(\cos\theta_2 + i\sin\theta_2)$, then:

$$z_1 z_2 = |z_1||z_2|\big(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)\big)$$

Two rules

Absolute values multiply: $|z_1 z_2| = |z_1| \cdot |z_2|$

Arguments add: $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

This is the polar-form version of "rotation + scaling."

De Moivre's Formula

Applying the multiplication rule n times:

$$z^n = |z|^n\big(\cos(n\theta) + i\sin(n\theta)\big)$$

De Moivre's Formula

Applying the multiplication rule n times:

$$z^n = |z|^n\big(\cos(n\theta) + i\sin(n\theta)\big)$$

Special case: If $|z| = 1$ (on the unit circle), then:

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$$

Raising to the n-th power = multiplying the angle by n.

Roots of Unity

Question: Which complex numbers satisfy $z^n = 1$?

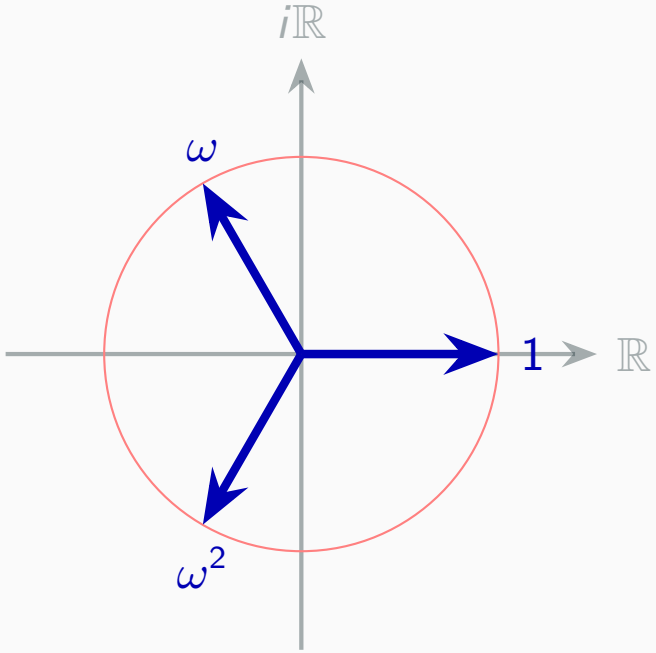
Roots of Unity

Question: Which complex numbers satisfy $z^n = 1$?

By De Moivre: need $|z|^n = 1$ and $n\theta = 2\pi k$ for some integer k.

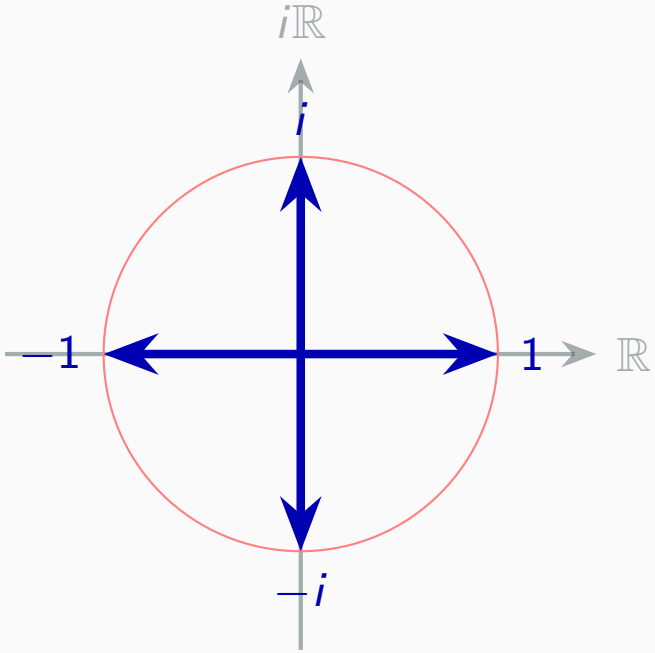
So $|z| = 1$ and $\theta = \frac{2\pi k}{n}$, giving n equally spaced points on the unit circle.

3rd roots of unity



$$1 + \omega + \omega^2 = 0$$

4th roots of unity



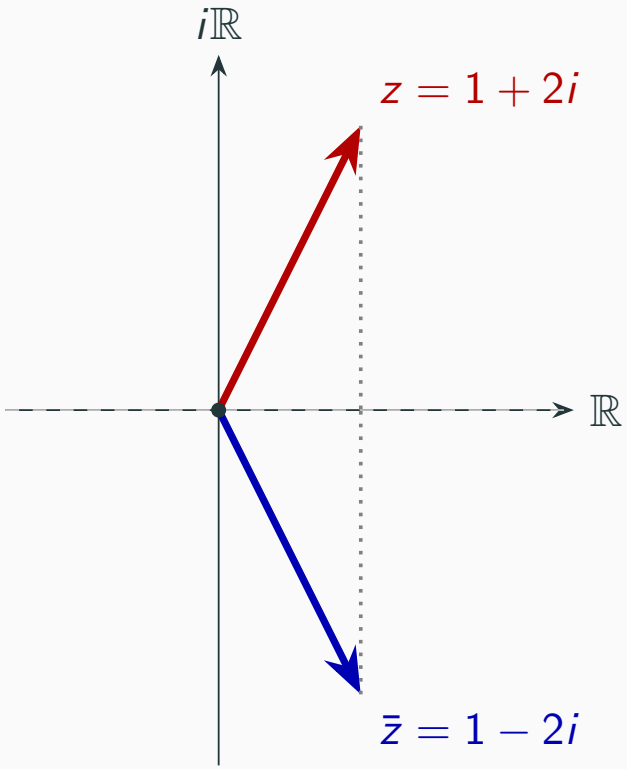
$$1 + i + i^2 + i^3 = 0$$

Conjugate, Division & Symmetry

Conjugate

For $z = a + bi$, the **conjugate** is:

$$\bar{z} := a - bi$$



Reflection across the real axis. Same real part, opposite imaginary part.

Properties of Conjugation

Key identity:

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2 \qquad \text{(always real, always} \geq 0\text{)}$$

Properties of Conjugation

Key identity:

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2 \qquad \text{(always real, always} \geq 0\text{)}$$

Extracting parts:

$$\text{Real part: } a = \frac{z + \bar{z}}{2}, \qquad \text{Imaginary part: } b = \frac{z - \bar{z}}{2i}$$

Properties of Conjugation

Key identity:

$$z\bar{z} = (a+bi)(a-bi) = a^2 + b^2 = |z|^2 \qquad \text{(always real, always} \geq 0\text{)}$$

Extracting parts:

$$\text{Real part: } a = \frac{z+\bar{z}}{2}, \qquad \text{Imaginary part: } b = \frac{z-\bar{z}}{2i}$$

Division: To compute $\frac{3+4i}{2+i}$, multiply top and bottom by the conjugate of the denominator:

$$\frac{3+4i}{2+i} = \frac{(3+4i)\overline{(2+i)}}{(2+i)\overline{(2+i)}} = \frac{(3+4i)(2-i)}{|2+i|^2} = \frac{10+5i}{5} = 2+i$$

What is Symmetry?

In mathematics, **symmetry** means **invariant under a certain action**.

The action on $\mathbb{C}$: **conjugation** $z \mapsto \bar{z}$.

What is Symmetry?

In mathematics, **symmetry** means **invariant under a certain action**.

The action on $\mathbb{C}$: **conjugation** $z \mapsto \bar{z}$.

Which numbers are **symmetric** (invariant)?

$$\bar{z} = z \quad \Longleftrightarrow \quad a - bi = a + bi \quad \Longleftrightarrow \quad b = 0 \quad \Longleftrightarrow \quad z \in \mathbb{R}$$

The **real numbers** are exactly the symmetric elements of $\mathbb{C}$ under conjugation.

What is Symmetry?

In mathematics, **symmetry** means **invariant under a certain action**.

The action on $\mathbb{C}$: **conjugation** $z \mapsto \bar{z}$.

Which numbers are **symmetric** (invariant)?

$$\bar{z} = z \quad \Longleftrightarrow \quad a - bi = a + bi \quad \Longleftrightarrow \quad b = 0 \quad \Longleftrightarrow \quad z \in \mathbb{R}$$

The **real numbers** are exactly the symmetric elements of $\mathbb{C}$ under conjugation.

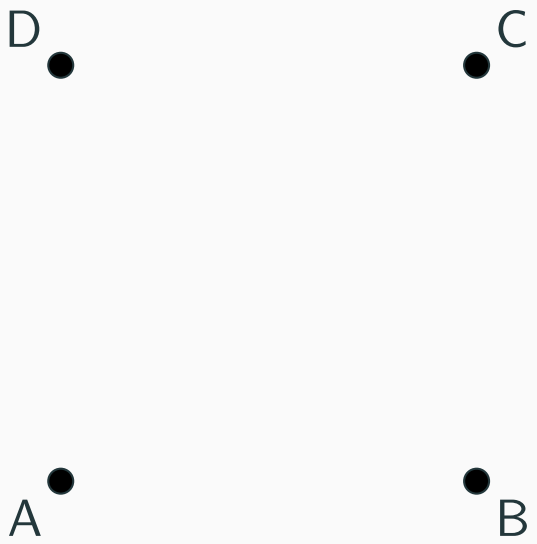
The Symmetry Principle

If a **symmetric system** produces an **asymmetric result**, then performing the symmetry action on that result produces **another valid result**.

Collecting all such results **recovers the symmetry**.

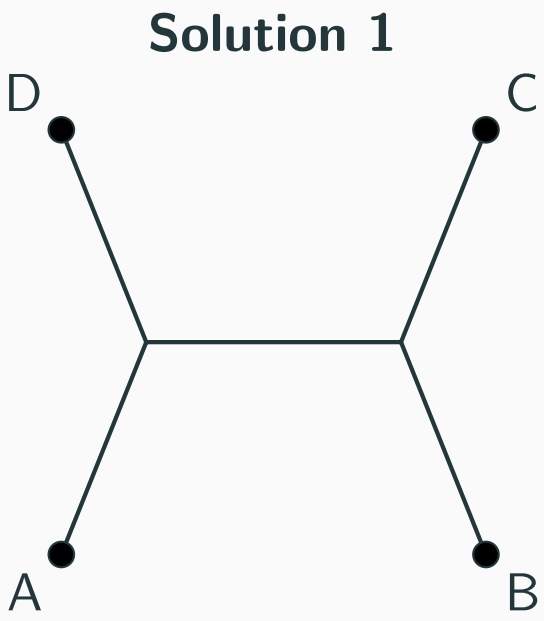
Example: Four Villages

Four villages at the vertices of a square. The system is **symmetric under 90° rotation**.



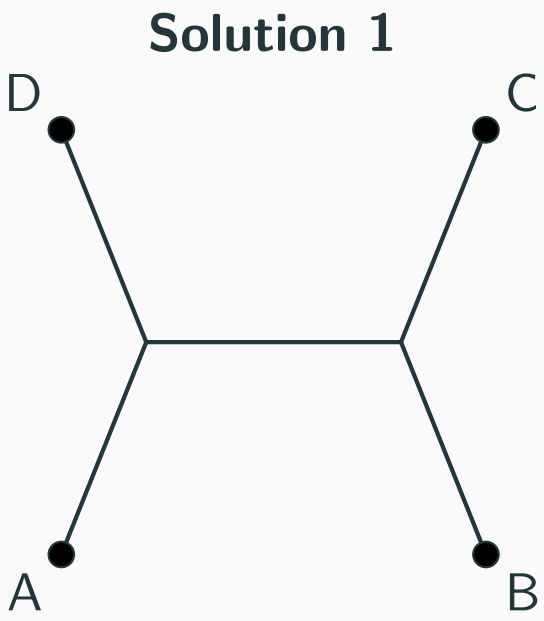
Problem: Find the **shortest road network** connecting all four villages.

Asymmetric Solution



This shortest road is **asymmetric** — it is NOT invariant under 90° rotation.

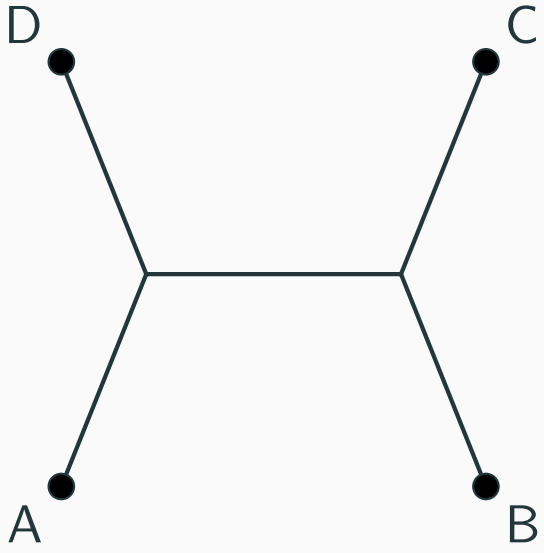
Asymmetric Solution



This shortest road is **asymmetric** — it is NOT invariant under 90° rotation.
But the problem IS symmetric!
So the symmetry action (90° rotation) applied to this solution must give **another valid shortest road**.

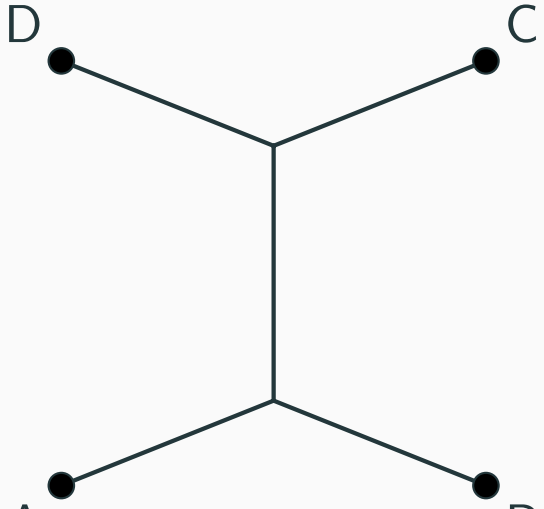
Asymmetric Solution

Solution 1



This shortest road is **asymmetric** — it is NOT invariant under 90° rotation.
But the problem IS symmetric!
So the symmetry action (90° rotation) applied to this solution must give **another valid shortest road**.

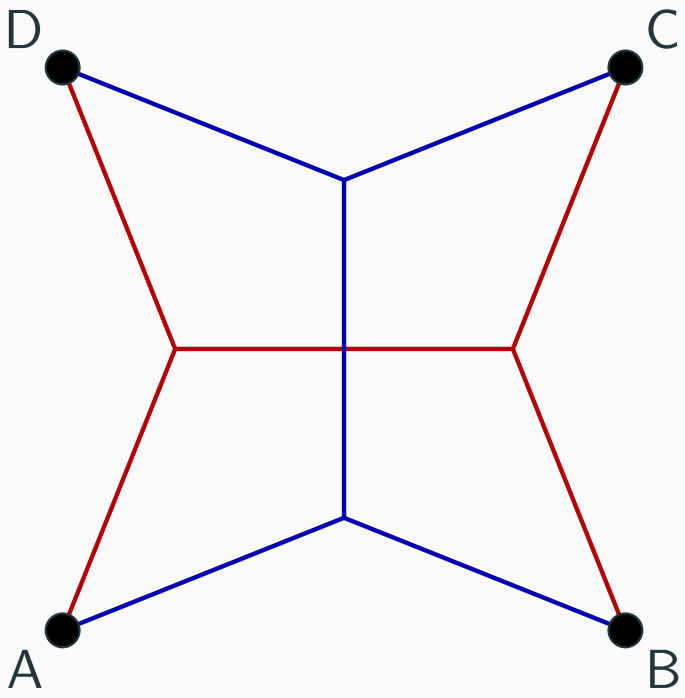
Solution 2 (rotated by 90°)



Also a shortest road. Also asymmetric.

Symmetry Recovered

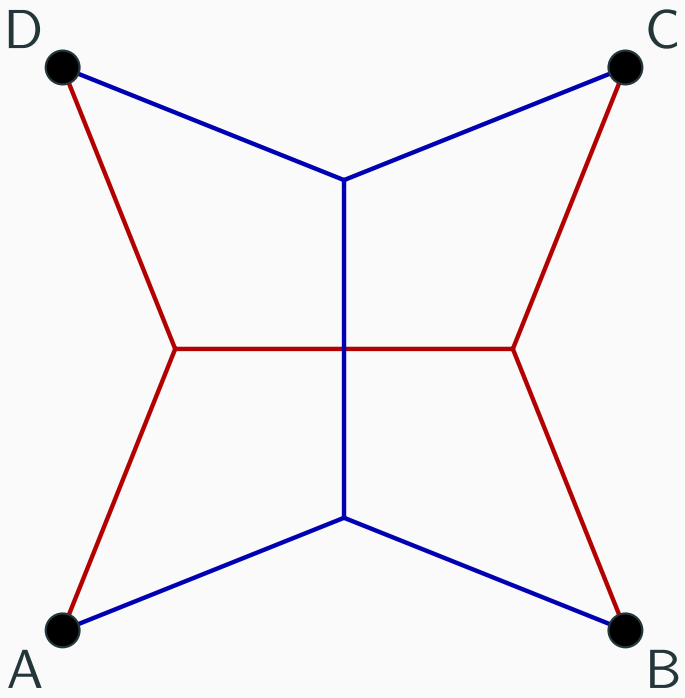
Overlay **both** solutions:



The combined figure **is symmetric** under 90° rotation!

Symmetry Recovered

Overlay **both** solutions:



The combined figure **is symmetric** under 90° rotation!

Individual solutions break the symmetry. The collection of all solutions recovers it.

Symmetry Principle Applied to Complex Numbers

A real polynomial is a symmetric system — it is invariant under conjugation.

If a real polynomial has a complex root z, then $\bar{z}$ must also be a root.

Symmetry Principle Applied to Complex Numbers

A real polynomial is a symmetric system — it is invariant under conjugation.

If a real polynomial has a complex root z, then $\bar{z}$ must also be a root.

$$t^2 - 2t + 5 = 0 \quad \Longrightarrow \quad t = 1 + 2i \text{ and } t = \overline{1 + 2i} = 1 - 2i$$

Symmetry Principle Applied to Complex Numbers

A real polynomial is a symmetric system — it is invariant under conjugation.

If a real polynomial has a complex root z, then $\bar{z}$ must also be a root.

$$t^2 - 2t + 5 = 0 \quad \Longrightarrow \quad t = 1 + 2i \text{ and } t = \overline{1 + 2i} = 1 - 2i$$

Complex roots of real polynomials come in conjugate pairs

If $p(t)$ has real coefficients and $p(z) = 0$, then $p(\bar{z}) = 0$.

This is the **same philosophy** as the four villages: the real polynomial is the symmetric problem, the complex root is the asymmetric solution, and conjugation produces the partner.

Symmetry of Eigenstructure

If A is a **real matrix**, then $\det(tI - A)$ is a **real polynomial**.

So eigenvalues come in conjugate pairs:

$$\text{If } \lambda = a + bi \text{ is an eigenvalue, so is } \bar{\lambda} = a - bi.$$

Symmetry of Eigenstructure

If A is a **real matrix**, then $\det(tI - A)$ is a **real polynomial**.

So eigenvalues come in conjugate pairs:

$$\text{If } \lambda = a + bi \text{ is an eigenvalue, so is } \bar{\lambda} = a - bi.$$

And the **eigenvectors** also come in conjugate pairs:

$$\text{If } A\mathbf{v} = \lambda\mathbf{v}, \text{ then } A\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}}.$$

Symmetry of Eigenstructure

If A is a **real matrix**, then $\det(tI - A)$ is a **real polynomial**.

So eigenvalues come in conjugate pairs:

$$\text{If } \lambda = a + bi \text{ is an eigenvalue, so is } \bar{\lambda} = a - bi.$$

And the **eigenvectors** also come in conjugate pairs:

$$\text{If } A\mathbf{v} = \lambda\mathbf{v}, \text{ then } A\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}}.$$

The symmetry principle applies to the entire eigenstructure

For real matrices: eigenvalues, eigenvectors, and spectral projections all respect conjugation.

This is not a coincidence — it is a direct consequence of the symmetry of real coefficients under conjugation.

Fundamental Theorem & Eigenvalues

Fundamental Theorem of Algebra

Theorem (Fundamental Theorem of Algebra)

Every degree-n polynomial with complex coefficients has exactly n roots in $\mathbb{C}$ (counted with multiplicity):

$$p(t) = (t - z_1)(t - z_2) \cdots (t - z_n), \qquad z_i \in \mathbb{C}$$

Fundamental Theorem of Algebra

Theorem (Fundamental Theorem of Algebra)

Every degree-n polynomial with complex coefficients has exactly n roots in $\mathbb{C}$ (counted with multiplicity):

$$p(t) = (t - z_1)(t - z_2) \cdots (t - z_n), \qquad z_i \in \mathbb{C}$$

Every polynomial factors into linear factors over $\mathbb{C}$.

No exceptions. No irreducible quadratics. No polynomials that "don't factor."

The Pipeline Works Again

For **any** $n \times n$ matrix A:

$\det(tI - A)$ is degree n $\quad\Longrightarrow\quad$ it has n complex roots.

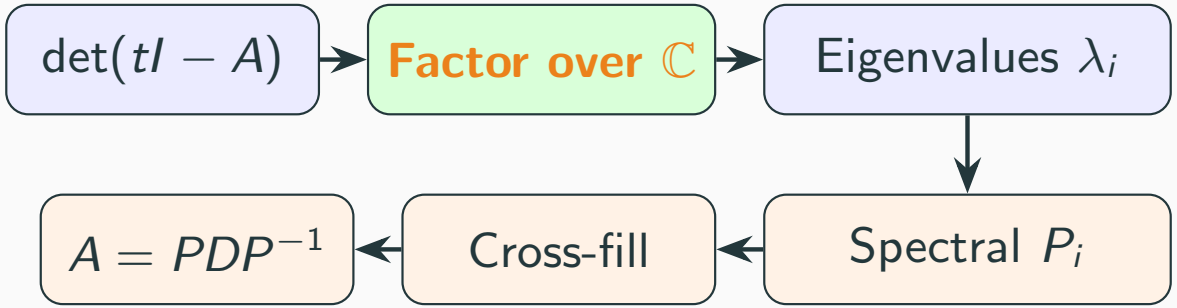
Every matrix has eigenvalues over $\mathbb{C}$.

The Pipeline Works Again

For **any** $n \times n$ matrix A:

$\det(tI - A)$ is degree n $\quad\Longrightarrow\quad$ it has n complex roots.

Every matrix has eigenvalues over $\mathbb{C}$.



The "Factor" step **never fails** over $\mathbb{C}$. The pipeline runs for all matrices with distinct eigenvalues.

Example 1: The Rotation Matrix Revisited

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \det(tI - R) = t^2 + 1$$

Over $\mathbb{R}$: **can't factor**. Pipeline stops.

Example 1: The Rotation Matrix Revisited

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \det(tI - R) = t^2 + 1$$

Over $\mathbb{R}$: **can't factor**. Pipeline stops.

Over $\mathbb{C}$:

$$t^2 + 1 = (t - i)(t + i)$$

Eigenvalues: $\lambda_1 = i$, $\lambda_2 = -i$. **Conjugate pair**, as the symmetry principle predicts.

Example 1: The Rotation Matrix Revisited

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \det(tI - R) = t^2 + 1$$

Over $\mathbb{R}$: **can't factor**. Pipeline stops.

Over $\mathbb{C}$:

$$t^2 + 1 = (t - i)(t + i)$$

Eigenvalues: $\lambda_1 = i$, $\lambda_2 = -i$. **Conjugate pair**, as the symmetry principle predicts.

Two distinct eigenvalues $\implies$ the pipeline works:

$$P_i = \frac{R - (-i)I}{i - (-i)} = \frac{R + iI}{2i}, \qquad P_{-i} = \frac{R - iI}{-2i}$$

$$R = i \cdot P_i + (-i) \cdot P_{-i}$$

Example 2: Eigenvalues $1 \pm 2i$

$$A = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}, \qquad \det(tI - A) = t^2 - 2t + 5$$

Over $\mathbb{C}$:

$$t^2 - 2t + 5 = \big(t - (1 + 2i)\big)\big(t - (1 - 2i)\big)$$

Example 2: Eigenvalues $1 \pm 2i$

$$A = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}, \qquad \det(tI - A) = t^2 - 2t + 5$$

Over $\mathbb{C}$:

$$t^2 - 2t + 5 = \big(t - (1 + 2i)\big)\big(t - (1 - 2i)\big)$$

Eigenvalues: $1 + 2i$ and $1 - 2i$. Conjugate pair.

Eigenvectors: $\begin{pmatrix} 1 \\ i \end{pmatrix}$ for $\lambda = 1 + 2i$, $\begin{pmatrix} 1 \\ -i \end{pmatrix}$ for $\lambda = 1 - 2i$.

Conjugate pair of eigenvectors for conjugate pair of eigenvalues.

Example 2: Eigenvalues $1 \pm 2i$

$$A = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}, \qquad \det(tI - A) = t^2 - 2t + 5$$

Over $\mathbb{C}$:

$$t^2 - 2t + 5 = \big(t - (1 + 2i)\big)\big(t - (1 - 2i)\big)$$

Eigenvalues: $1 + 2i$ and $1 - 2i$. Conjugate pair.

Eigenvectors: $\begin{pmatrix} 1 \\ i \end{pmatrix}$ for $\lambda = 1 + 2i$, $\begin{pmatrix} 1 \\ -i \end{pmatrix}$ for $\lambda = 1 - 2i$.

Conjugate pair of eigenvectors for conjugate pair of eigenvalues.

The symmetry principle in action: real matrix $\implies$ everything conjugates.

What We Have Achieved

Restriction (a): Resolved

Over $\mathbb{C}$, every polynomial factors into linear factors.

Our pipeline works for all matrices whose minimal polynomial has **distinct roots** — regardless of whether eigenvalues are real or complex.

What We Have Achieved

Restriction (a): Resolved

Over $\mathbb{C}$, every polynomial factors into linear factors.

Our pipeline works for all matrices whose minimal polynomial has **distinct roots** — regardless of whether eigenvalues are real or complex.

Restriction (b): Still open

Repeated factors in the minimal polynomial $\implies$ Lagrange interpolation fails (observation points collapse).

This requires **Jordan Canonical Form** — self-study material.

Complex numbers do not solve this.

Preview

What We Have Gained

Every matrix with distinct eigenvalues (over $\mathbb{C}$) is diagonalizable:

$$A = PDP^{-1}$$

Spectral decomposition, **value tables**, A^n — all work with complex eigenvalues.

The rotation matrix R with eigenvalues $\pm i$, the matrix A with eigenvalues $1 \pm 2i$ — our full machinery applies.

Next Lecture: The Exponential Function

We can compute $A^n = PD^nP^{-1}$. But what about e^A?

Next Lecture: The Exponential Function

We can compute $A^n = PD^nP^{-1}$. But what about e^A?

If $A = PDP^{-1}$, then we would want:

$$e^A = Pe^DP^{-1} = P\begin{pmatrix} e^{\lambda_1} & \\ & e^{\lambda_2} \end{pmatrix}P^{-1}$$

Next Lecture: The Exponential Function

We can compute $A^n = PD^nP^{-1}$. But what about e^A?

If $A = PDP^{-1}$, then we would want:

$$e^A = Pe^DP^{-1} = P\begin{pmatrix} e^{\lambda_1} & \\ & e^{\lambda_2} \end{pmatrix}P^{-1}$$

The question: What does e^λ mean when λ is **complex**?

What is $e^{i\theta}$? What is e^{1+2i}?

Next Lecture: The Exponential Function

We can compute $A^n = PD^nP^{-1}$. But what about e^A?

If $A = PDP^{-1}$, then we would want:

$$e^A = Pe^DP^{-1} = P\begin{pmatrix} e^{\lambda_1} & \\ & e^{\lambda_2} \end{pmatrix}P^{-1}$$

The question: What does e^λ mean when λ is **complex**?

What is $e^{i\theta}$? What is e^{1+2i}?

Next lecture: We will discover that $e^{i\theta} = \cos\theta + i\sin\theta$, and use this to solve systems of differential equations $\mathbf{y}' = A\mathbf{y}$.